Stock Return Forecasting Model of Top UK Technical Firms : Investors Model for Investment Decision-Making

Using Advanced Forecasting Techniques and Advance Machine Learning Algorithm

(https://github.com/Thesineo/sincos)

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Abstract

The rapid growth of the UK Tech sector in the past few years has captured major attention of investors to capitalise on its potential. Making accurate investment decisions is crucial for investors to attain maximum returns from these tech sectors. This growth of Tech sectors is governed from the stock movements of the technical firms registered on the FTSE TechMark index at London Stock Exchange. Accurate forecasting of returns for these technical stocks are essential for investors to make precise investment decisions in this sector. This study aims to comparatively analyse the performance of Autoregressive Integrated Moving Average (ARIMA) and General Autoregressive Conditional Heteroscedasticity (GARCH) models to forecast the stock return values of top technical firms in UK, providing an assist to investors to make accurate investment decisions in concern to the Technological sector. Both the forecasting models are analysed for a period of 7 years, starting from January 01, 2015 to December 31,2022. For the chosen 10 technical firms, both the fit models are estimated on the basis of their log-likelihood values, Akaike Information Criterion and Bayes Information Criterion Values. Later, they are evaluated on the terms of forecasting measures such as Root Mean Square Error (RMSE) and Mean Absolute Percentage Error (MAPE). ARIMA models tend to estimate different orders for every individual dataset, whereas GARCH models induce a single model for all the considered stock movements . In GARCH, the asymmetric GJR-GARCH(1,1) model outperforms other GARCH models by producing low information criterion values and high log-likelihood values. Overall, the study recommends that the GJR-GARCH(1,1) model better fits the stock movements dataset of technical firms to forecast the stock return values in UK in comparison to the ARIMA models, making it an asset for investors to make accurate investment decisions on the growing UK technological sector .

1.Introduction

1.1 Forecasting Stock Returns

Hypothetical and experimental research have shown us that stock markets and economic upturns are correlated in a positive manner(Kim et al.,2011). As stock markets are pervaded by their dynamic, complex and volatile nature, investment decisions play a significant role in manipulating the stock movements in the markets through their prior occurring forecasting values. Signifying the importance of forecasting, the primary concern for this process would be analysing the predictive forecasting models for respective stock movements, followed by their stock return values. Following this issue, many researchers have paid serious attention towards increasing the model accuracy for the prediction of stock movements, followed by the estimation of their returns. Depending on these forecasting models, investors and investment institutions will be dynamic and transcend in their decision-making abilities to maximise their returns on their investments in the long run. If a predictive forecasting model can provide a good estimation of returns for their investment, by forecasting a sequence of stock movements prices,then it would minimise the risk involved in the process. Which would help the investors to make discrete decisions on their investments and maintain the efficiency of their portfolios, in the stock market. Regardless of the uncertainty conditions, being controlled by applying accurate forecasting tools in accordance to the stock market strategies (Karaca et al, 2019) .

1.2 UK Tech Sector

UK in recent years has emerged as one of the influential countries in becoming a global landscape for technological innovation, making a huge impact on its economic growth. Apart from being globally influential , the UK tech sector has also retained its status for raising the highest venture capital investment in the European continent with a market value capitalisation of £860 billion, being third largest in the world behind the technological sectors of US(£15.7

trillion) and China (£4.2 trillion). The statistics of this sector in the UK say that , for the six years span starting from 2015 technology sector showed good figures of hope to grow the UK economy in comparison to other wider sectors in the country following positive trends for the period. In 2019, the UK emerged as the fifth biggest technological service exporter in the world, rising from 25% in 2015, showing the capability to attract international investors in the sector to 63 % in 2020 increasing from 45% in 2015 (Middleton,2021). In 2020, UK showed the highest rate of growth by increasing its tech investments by 17 % in comparison to its previous years. Later being struck by ruthless factors of COVID-19 the sector managed to stabilise its condition in a quick manner to extensively support the economy during the pandemic .Exhibiting its resilient nature the technological sector governed to grow in spite of being struck by pioneering circumstances of the pandemic, such as the Brexit transition period and lockdowns (Middleton, 2021). Above this , the UK Tech sector played a major role in supporting the economic growth and employment issues during the pandemic to adapt to the changes of COVID-19 in the country(Middleton, 2021). Considering the future growth of this sector , investments on the stocks of technological companies are important to generate at most returns and revenue from these tech stocks to not only help the economic growth but also to influence the growth of market capitalisations . Computing the importance of investment in the UK Tech Sector it is also important to throw light on the investment decisions taken on these technological companies on the basis of the forecasting methods.

1.3 Investors Decision making through Forecasting

Stocks display an influential characteristic on the growth and performance of these technological firms on the basis of their innovation , financial stability and investors confidence. Stock markets being one of the important aspects for the growth of the economy and various sectors, stocks of individual companies govern their confidence of motion for investors to make heavy investment in their respective stocks for future growth of the companies . Investors in this process make use of various analytical tools and strategies to confirm their guaranteed returns before making any investments. One of them being investing in growth stocks, which are the

stocks of the companies that are expected to increase their returns at a faster pace than the average time. But these investments cannot guarantee to make extensive returns for the investors as these stocks may involve volatility in their stocks movements, due to high expectations of investors to constantly manipulate their performance in comparison to other stocks. To assess these volatile ranges of stocks , investors in their analysis make use of different methods to predict the future stock return prices and the volatility in their movements to make a safe and guaranteed investment .

Investors for their analysis rely on many tools , but previous studies have shown that forecasting methods have proven to accurately predict the stock returns and volatility for many stock markets and companies, being a reliable asset for investors in making their investment decision (Agarwal et al.,2013). Governing the rising growth of UK tech sectors and the importance of Forecasting methods for investment decision making in the given time span, the study helps to comparatively assist the investors with the best forecasting method for predicting the stock returns of these technical firms falling under the UK tech sectors. Since 2015 , due to flourishment of the UK Tech sector many Tech IPOs were listed on the London Stock Exchange ,making a minimum of 5 billion market capitalization by each technological firm termed as blue chip Tech stocks in the stock market .

1.4 FTSE techMark Series

London stock exchange being one of the oldest stock exchanges in the world creating a pathway for growth , to help the companies raise their capital and investors to build their portfolios across a range of sectors in the global market. Furthermore, the study makes use of FTSE techMARK index series which is a series computing of technological stocks listed on the London Stock Exchange's market as classified by the Industrial classification benchmark (ICB) , which provides the investors a tool to gauge the performance of the companies coming under the technological sector(*FTSE techmark index series* , 2023). Confronting the growth of UK Tech sectors in coming years , we use 10 of the technical firms from the sector on the basis of their

market capitalisation listed on the FTSE 100 techMark series index , making it a platform to govern the past volatility , stock movements and past historical prices to forecast the future stock returns of this firms by comparatively analysing the two most efficient forecasting methods for investors to make effective decisions while making an investment.

Company Name	Market Capitalisation (GBP)	Description of the Company
The Sage Group PLC (SGE.L)	9.41 Billion	Software Solutions
Smith and Nephew PLC (SN.L)	10.31 Billion	Medical technology Company
BAE Systems (BA.L)	27.98 Billion	Defence Technology
Computacenter PLC (CCC.L)	2.48 Billion	Technological Service Provider
Genus PLC (GNS.L)	1.58 Billion	Genetic Technology
QinetiQ Group (QQ.L)	1.98 Billion	Technology Product Services
Flutter Entertainment (FLTR.L)	28.79 Billion	Online Betting Company
Renishaw PLC (RWS.L)	2.90 Billion	Scientific Technology Company
Spirent Communication PLC (SPT.L)	1.01 Billion	Communication Technology Services
Spectris PLC (SXS.L)	3.79 Billion	Instrument Technology

Table 1.1 Companies listed on FTSE techMark for analysis (*FTSE techmark 100: Market overview* ,2023)

1.5 Objectives and Forecasting Models

The main motive of forecasting is to support the investment decisions to enhance the investors performance efficiency. Many studies have been published which aim to find the best forecasting models to accurately predict stock returns in the most rapid way (Javier et al.,2003) .

Concerning the financial time series , the Autoregressive Integrated Moving Average (ARIMA) model has proved to be one of the most efficient methods for investors in forecasting accurate predictions regarding figures of stocks. While forecasting the returns for S&P BSE IT returns by ARIMA model depicted that investors can make their investments on the basis of forecasted returns predicted by ARIMA model to make upstanding portfolios of profitable stocks (Chall et al.,2020). Governing the same financial time series , some stocks may implement the volatility nature in their movement and to accurately forecast such kinds of stocks investors gradually lean towards the Generalised Autoregressive Conditional Heteroscedasticity (GARCH) models. Dealing with the volatile Nigerian stock market, author infers that GARCH models provide a better robust forecast for the returns for investors to make their decisions by minimising the risk involved through the factor of volatility in stock movements (Atoi , 2014). Therefore, a comparative analysis is required for both the forecasting models to check the better fit model that can predict the stock returns of the technical firms depicted above, to provide a supportive method for investors to make evident decisions on their investment compiling the required time horizons to increase their profits.

The main hypothesis regulated by the study is that for volatile or nonvolatile financial time series, models exhibit a normal distribution for their residuals by producing smaller magnitudes of measuring values to accurately forecast the returns for different time horizons in comparison to the other model . The empirical results of both the models will be tested on the urge of providing a normal distribution for there residuals on the basis of there accuracy parameters to forecast the stock returns for concerned time horizons , such as MAPE (Mean Absolute Percentage Error) , RMSE (Root Mean Square Error). The models will be analysed for long term (Multi-step forecasts) horizons on daily observation of stock movement data to give investors a wider spectrum to make their investments on their requirement of time span and rely on subsequent data types for their investments. These models can relatively gauge the short term predictions made through changing the time horizons . Apart from daily observed stock data these models will also be efficient to forecast the returns on the basis of weekly and monthly stock movements datasets for the investors to have a broad spectrum to make accurate investment decisions. Next section describes a detailed investigation on the vast content of

literature on the need of Stock return forecasting and the methods used in the study for comparative analysis.

2. Literature Review

2.1 Forecasting of Stocks

Stock return forecasting processes are deemed to be one of the most important aspects in developing the area of investment decision making . Focusing on predicting stock prices , analysis can be segregated into two components - Fundamental and Technical analysis. The Fundamental Forecast analysis is carried out on the basis of financial attributes of the company , whereas the technical forecast analysis uses historical data to forecast the future prices of the stocks on the assumption that the market trends will be repeating itself (Levy,1967). This theory became one of the statistical strategies for the component of investment decision making , but later it was opposed by the efficient market hypothesis which stated that past prices cannot predict the future stock prices (Fama,1970). In 1987, evidence was found that the predictable volatility of the stock returns is positively related to the expected market risk premium and negatively related to the unexpected market risk premium (French et al.,1987) . Following him , Chou in 1988 using the same source of data of NYSE stock index from January 1928 to December 1984 supported French et al. 's by finding about the positive relation between stock return and volatility of their predictability components (Chou , 1988) . On the other hand, studying the same source of data , contradicting the statements of French et al. and Chou it was concluded that any relationship stated between variance and returns are proved to be weak (Baillie and DeGennaro,1990).

Depending on this, Welch and Goyal (2008) with the help of compound returns on S&P 500 index forecasted the equity premium, which covered the dividends and rate of treasury bills by establishing a link between forecasted and real economic values(Welch and Goyal, 2008). In 2020, a study generalised on forecast dynamics of returns and volatility for S&P BSE index for

investment decision making stated that the forecasted values for returns are almost equal to the actual economic values adding more value to the forecasting model to predict in medium and long term horizons(Challa et al.,2020). Neely et al. (2014) governed the forecast of stock returns with the help of technical indicators stating that these are one of the most significant indicators economically and statistically. Furthermore, based on US stock returns a dynamic relation was stated between the volume of stocks and the returns which showed that investors do not attain much profits from the volume curve (Wang et al.,2018). An experiment conducted in 2018 to forecast the oil prices by using 18 macroeconomics and 18 technical indicators showed the accuracy of forecasts by generating a certain return profits for mean-variance investors (Zhang et.al,2018). In 2019, Zhang along with the other authors explained evidence of the U-shaped investment curve along with the trading behaviour of intraday stock movements , which resulted in computing significance of afternoon stock return prediction using the morning returns (Zhang et al., 2019).

2.2 ARIMA Model

The existing body makes use of the Autoregressive Integrated Moving Average (ARIMA) model which is known as the efficient model for investors to forecast the future stock prices. Box and Jenkins introduced this model in 1976 making a new framework for forecasting (Harvey and Todd, 1983). The significance of the model was on analysing the stochastic properties of economics rather than relying on constructing congruent equation models where the data would speak for itself(Gujarati ,2022). This model was used to forecast the stock values in various stock markets . An analysis done on the Indian Stock Market for six years presented a time series ARIMA model (1,0,1) which helped the investors and institutions to predict the future indices of the stock market (Banerjee,2014). This model satisfied the conditions for the goodness of fit than other forecasting models being stationary and invertible . Understanding the importance of forecasting methods in stock returns, few studies encouraged the ARIMA model as it was capable of producing superior forecasting time series for the stock returns (Adebiyi and Oluinka ,2014). Following this a study conducted on identifying the best models to minimise the risks for investors while making an investment in stocks found out that ARIMA model outperformed many artificial intelligence and frequency domain models in accurately predicting the stock

returns for specified companies (Mallikarjuna and Rao ,2019). Apart from this ARIMA models were also used to analyse the stability of the markets depending on the type of the data . With concern to the London Stock Exchange a study was carried out to test the financial stability and the market efficiency between S&P 500 and LSE for time series forecasting of stock returns on the basis of monthly and yearly data(Rounaghi and Zadeh , 2016). This research resulted in the findings that the S&P 500 ARIMA model outperformed the London stock exchange in forecasting stock returns by being capable of predicting real values for medium and long horizons (Rounaghi and Zadeh , 2016) . This investigation also showed that the ARIMA model with monthly stock returns outperforms the yearly returns by providing more financial stability to the stock markets (Rounaghi and Zadeh , 2016).

In a similar manner , the ARIMA model was used for S&P BSE sensex to forecast the beta values for top companies performing impeccably on the index due to their beta value. The results in this study showed that the estimation of the model was completely acceptable as the forecasted value accurately correspond to the actual beta values of the company, making the model insightful for investment decisions by connecting the forecasted beta values to their respective stock investments and returns (Challa et al., 2018). A case investigated on the BRIC (Brazil , Russia, India, China) countries to establish the relations between their macroeconomic variables for stock returns used the ARIMA model to find the significant association of their macroeconomic factors. The result later showed that no statistical relation was established between the factors of these given countries and their stock returns , indicating each country specifies and governs to their own forecasted return rather than being independent on other countries (Gay,2016).

Apart from the use of the ARIMA model in forecasting stock returns in the stock market this model has been put to use in various other cases. Guha in 2016 used this ARIMA model to forecast the gold prices in India which helped the investors to clearly picturize the events about when to buy the gold and when to sell it (Guha and Bandyopadhyay , 2016). The ARIMA model was infused during the COVID-19 period to forecast the number of cases per 1 million , to fight against the global threat (Hernandez et al.,2020). This case study was carried out by proposing a

relationship between countries depending on their geographic region or areas, to predict the advancement of viruses among people. The results of this study showed that the ARIMA model was able to predict the COVID-19 cases with the RMSE averages concerned to specific regions (Hernandez et al., 2020). The ARIMA model was also aimed to forecasting the Irish inflation concerned to the central banks of ireland to emphasis there economic conditions , here the study was aimed more towards minimising the out-of-sample forecast errors for the values predicted by the ARIMA model than on maximising the in-sample goodness of fit (Meyler et al.,1998) . ARIMA displayed its efficiency in other fields also, by forecasting the future prices for crude oil prices making it one of the important aspects of study for the human race . ARIMA (1,1,0) was used to forecast the oil prices in Ghana in a stochastic format due to the growing demand in petroleum in this region (Abledu and Kobina , 2012) . In a similar manner for oil prices in Nigeria , ARIMA (2,1,2) was challenged by ARIMA (6,1,7) on the basis of ACF and PACF plots but on the terms of parsimony the author stood still with the displayed model of ARIMA(2,1,2) (Akpanta and Okorie,2014).

Conceiving all the advantages and efficiency of the ARIMA model , this model also deploys some limitations . In a study conducted on stock price prediction of selective stocks , the author points out that for short term forecasting of the values ARIMA model provides strong and vigorous results , but when dealing with high volatile cases for long term forecasting the model falls back in some cases (Poongodi et al.,2014). While dealing with financial time series for forecasting short and term and long term values , ARIMA might be a good fit for the data but implementing its limitation while dealing with data or stock movements with high volatility in its historical data and for long term assets the literature governs us towards more stochastic models such as Generalised Autoregressive Conditional Heteroscedasticity models .

2.3 GARCH Model

In comparison to the ARIMA model the study also influences work on the volatile estimation of the financial time series for an heterogeneity model of variance named as Generalised

Autoregressive Conditional Heteroscedasticity (GARCH) model . Depending on a study for heteroscedasticity , the author makes a statement of observing an autocorrelation of an autoregressive structure in a series of unequal variance over a range of different periods(Gujarati,2022) . Thus on this , a study by Engle (1982) introduced the Autoregressive Conditional Heteroscedasticity(ARCH) process for capturing the effective behaviour of the conditional variance using the lagged values for a time varying conditional model (Dana ,2016). On the basis of this study Bollerslev inferred the GARCH model , where the variance of the model for a period was dependable on both , the disturbance term and the conditional variance of the previous period (Abounoori & Tour ,2019). Later in the year , Akigray (1989) used the GARCH (1,1) model to forecast the volatility for the US stock markets monthly data and found out that GARCH model produces better forecasts in comparison to other forecasting tools (Akigray,1989). Many studies have undertaken the case of GARCH models to forecast and study the volatility function of their stock markets. While implementing the volatility to forecast the stock returns for Stock exchange of Hong Kong , author tried to test the ability of GARCH models and its components to capture the volatility and returns for 14 selective stocks registered on the basis of decomposition of their conditional variance to improve their goodness of fit for different time horizons (Ane , 2006) . The results showed that for short run time horizons the GARCH model provides more accurate forecasts , whereas for longer time horizons Components of the GARCH model were selected , as the traditional GARCH model forecasts deviated from its realised variances (Ane,2006). One such case was conducted on Ghana Stock Exchange , for forecasting the values and volatility of the returns on the stocks of GSE, where three GARCH models GARCH(1,1) , EGARCH(1,1) , TGARCH(1,1) along with Random walk (RW) where compared and tested to forecast the volatility returns. The outcomes of the study stated that on the basis of volatile clustering and leptokurtosis GARCH(1,1) outperformed every model by following a random distribution for residuals (Frimpong and Oteng-Abayie,2006).

GARCH model was also used to study the monthly stock returns of the market in seven emerging countries , where it showed a comparison of linear and non-linear GARCH models (Gokcan,2000) .This study later disclosed that linear GARCH (1,1) model was efficient in forecasting the volatility of the stocks market returns than the nonlinear EGARCH

model(Gokcan,2000). Contradicting this study , a research on volatility of daily stock returns for Total Nigeria Plc used nine different variants of linear and non linear GARCH models to forecast the volatility of the returns with historical data of the past 17 years (Emenogu et al.,2020). This result disclosed that the nonlinear sGARCH and EGARCH models were better in terms of forecasting the volatility in comparison to the other models (Emenogu et al.,2020). Conditional distributions play a major role in forecasting of stock returns volatility when combined with the GARCH models , an analysis carried out on estimating the volatility of the Tel Aviv Stock exchange (TASE) proved that GARCH model with fat tailed densities improves the efficiency for measuring variance , which in turn indicates that the GARCH models with the skewed Student-t distribution are successful for forecasting future volatilities (Alberg et al.,2008). While some cases analysing the GARCH models under different distribution assumptions, suggest that during the process of investigating forecasting performance of the GARCH(1,1) model nine different distributions were checked for S&P 500 Index Future Returns (Wilhelmsson,2006). This investigation showed that for weekly and monthly data forecast horizons, skewness and variation in higher value of distribution do not increase the efficiency of the forecasts(Wilhelmsson,2006).

Apart from this , the GJR-GARCH(1,1) model during an study conducted on analysing the financial time series behaviour of daily stock returns for four firms listed on the Nigerian Stock exchange showed that it gave a better fit for the data and found to be superior for forecast evaluation of in and out samples in comparison to the normal GARCH(1,1) and EGARCH(1,1) models (Onwukwe et al., 2011). During a model comparison study between the GARCH,GARCH-M, GJR-GARCH, and log GARCH models to forecast the returns volatility on the basis of DJIA, S&P 500 and S&P CNX Nifty indices found out that the GJR-GARCH model provided a better fit for the data to forecast the volatility in comparison to the other models (Nugroho et al. , 2019). Understanding the ability of different GARCH type models to not only be a better fit for the financial time series data but to exhibit their efficiency in capturing their historical volatility to predict the future returns of the stocks along with their conditional variance . This study makes use of the literature and the different types of GARCH model to forecast the future stock returns of the firms on the basis of their conditional distribution and

variance. And later contrast the model's residual distribution with the ARIMA model for being a better fit governing specific firm stocks. Above this the literature also shows some hybrid nature of the GARCH model to work with ARIMA models to forecast returns. A study on forecasting the S&P 500 stock prices in the US stock market showed that , indulging a 2-step procedure on analysis, the ARIMA(2,1,2)-GARCH(1,1) model provided a better fit to forecast the S&P 500 stock prices. Above this GARCH models were also used in many other fields to forecast the prices (Mustapa et al.,2019). This GARCH model was used to forecast the commodity of agricultural crop prices such as sorghum , wheat and soybean in US , which showed that the specification of this model gave an merit to identify the asymmetrical distributed errors and provide a improved liable forecasts of this crop prices (Ramirez and Fadiga, 2003). Furthermore , researchers forecasting the volatility of crude oil prices stated that the GARCH model was more efficient in comparison to other forecasting models acknowledging the accuracy they provide for the future values (Tripathy and Rahman, 2013). Above this they also stated that the asymmetric GARCH models had an diverse effect in modelling the volatility for different concerned commodities among the other GARCH models

2.4 Comparative Analysis

The content of literature also shows the comparison between the ARIMA and GARCH models to specify the better forecasting model among the two models showcasing an efficient forecasting accuracy. A study concerned with forecasting crude oil prices to avoid future hikes for non-renewable sources found out that the GARCH(1,1) model was better fit to forecast the daily crude oil prices in comparison to ARIMA(1,2,1) model (Nian, 2009). In a similar manner while predicting the future gram prices in India for a specific market it was found out the GARCH model showed a better than the ARIMA model to forecast the future prices of the grams(Bharadwaj et al.,2014). A comparative performance was also stored while , testing the abilities of the ARIMA and GARCH models to forecast the time series for selective 31 stock log returns in the US stock market , whose results concluded that GARCH models were more efficient in the task as they provided a normal distributions for residuals where as the ARIMA

model failed to provide the distributions for some horizons (Sparks and Yurova,2006). Apart from this some cases like , forecasting the volatility for future Malaysian market properties rate and share price showed that ARIMA model provided a better fit for the forecasts in comparison to the GARCH model (Miswan et al,2014). Another similar comparative on predicting the crude oil prices during the COVID-19 pandemic effect showed that to increase the use of the oil to stabilise the major economic activities from Saudi Arabia showed that ARIMA(4,1,4) provided more accurate forecasts for future oil price in comparison of GARCH(1,1) model (Haque and Shaik,2021) . Similarly, a relative comparison considered in predicting short time traffic flow prediction in China showed that to control future traffic networks management ARIMA model provided more efficient forecasting results than GARCH (1,1) model (Chen et al.,2011). Concreting the content of literature , the subject tries to add , the comparative analysis of ARIMA and GARCH models for forecasting the stock returns of selective technological firms listed on FTSE techMark to provide investors an precise model to make effective investment decisions with the predicted forecasts . The next section gives detailed objectives of each model and data types used in the study

3.Data & Methodology

Data for the study was collected from yahoo finance (Yahoo Finance) for ten selective technical firms shown in Table 1 listed on the FTSE100 techMark in the London stock exchange. The empirical analysis was carried out on returns of the selective stocks for technical firms in R studios , from January 1, 2015 to December 31,2022. Furthermore, from the recorded 2021 daily observations, 1518 observations (January 1,2015 - December 31, 2020) were detained for training the model, whereas the remaining dataset of 503 daily observations (January 1,2021 - December 31,2022) were used to test the accuracy of the models on the basis of their forecasted values. Apart from this , the analysis also makes use of the weekly and monthly observations for the stock returns of these technical firms following the same time horizon to provide a

cumulative evaluation of the models for the investors decision making, helping them make accurate choice of the models for their concerned horizons of investments. For the empirical analysis of the investors' decision-making process, the study makes use of two extensive forecasting methods, ARIMA and GARCH models, for forecasting the stock return values for the ten selective technical stocks.

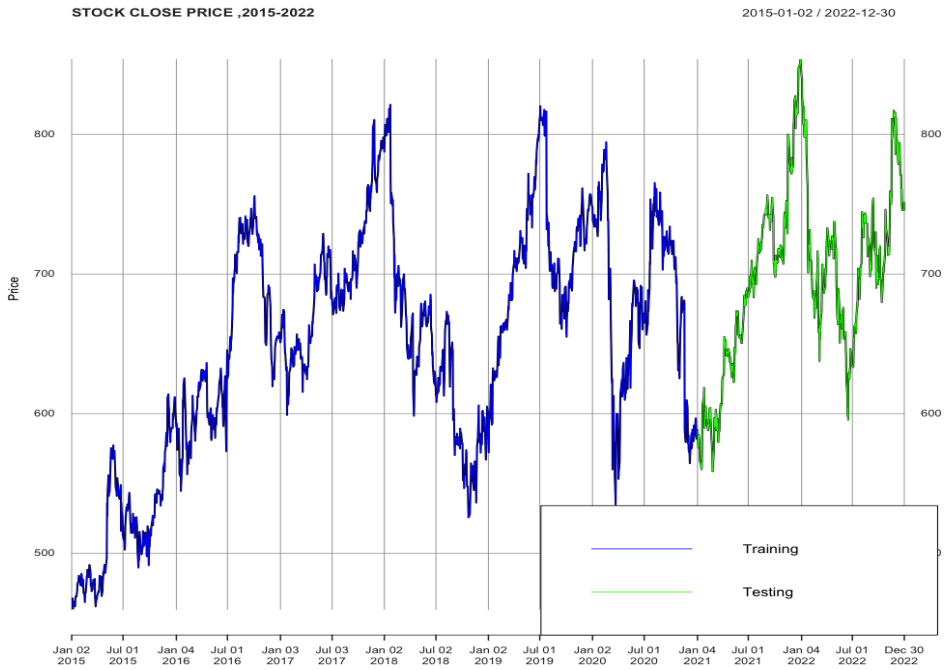


Fig 3.1 Training and Testing Dataset for the Analysis of the Model

Preliminary requirement of analysing both the financial time series forecasting models, is to use a stationary dataset to support the future performance of the process. A dataset is said to be stationary over a time interval if it provides constant values for mean, variance, and autocorrelation structures. To check the stationarity of the dataset , we make use of a statistical significance test called the Augmented Dickey-Fuller Test (ADF test) (Paparoditis and Politis,2018). Along with that, the test also provides the number of differencing required for the

dataset to convert it into a stationary time series . Depending on the p-value of the ADF test , the dataset can be transformed into a stationary time series by carrying out a first to 'n' differencing functions. Once the stationary of the time series has been achieved, further process of model fitting can be initiated.

3.1 ARIMA Method

Introduced by Box and Jenkins in 1970, ARIMA is one of the most popular linear function models used in forecasting financial time series . Making use of the linear functionality through some previous observations of the variables and computing random errors, the ARIMA model forecasts the future values of the variable.The general form of defining this time series is :

$$y_t = c + \phi_1 y_{t-1} + \phi_2 y_{t-2} \ldots\ldots\ldots, \phi_p y_{t-p} + \theta_1 \varepsilon_{t-1} + \theta_2 \varepsilon_{t-2} \ldots\ldots \theta_q \varepsilon_{t-q} + e_t \qquad (1)$$

where,

y_t is the time variable

c is the intercept value

ϕ_p and θ_q are the model parameters

e_t is the error term

ε_t are the random errors

This Box and Jenkins method can also be stated in the form of ARIMA (p,d,q) model where p is the number of autoregressive (AR) terms, d is the number of differencing needed for the dataset to be a stationary time series and q defines the number of moving averages (MA) terms for the model . Considering the ARIMA model keeps a constant variance of the data, for the ARIMA(p,d,q) model equation 1 can be represented as ,

$$(1-\phi_1 B - \ldots - \phi_p B^p)(1 - B)^d y_t = c + (1 - \theta_1 B - \ldots - \theta_q B^q) e_t \qquad (2)$$

where,

$1-\phi_1 B - \ldots - \phi_p B^p$ is the autoregressive (AR) compound of order p

$1 - \theta_1 B - \ldots - \theta_q B^q$ is the moving average (MA) compound of order q

$(1 - B)^d$ is the stationary compound of differencing order d

c is the constant term

e_t is the error term for the model

3.2 GARCH Method

Discovered by Bollerslev in 1986, GARCH model governs the differing variance in a financial time series defining the heteroscedasticity of the model. GARCH models applied to specific datasets tend to possess high volatility across the time series for financial conditions. This model can generally be written as GARCH (q,p) where q defines the number of moving average (MA) terms and p expresses the number of autoregressive terms in the model. Having a diverse variance, the model can be expressed as :

$$\sigma_t^2 = \omega + \Sigma_{i=1}^{p} \alpha_i \varepsilon_{t-i}^2 + \Sigma_{j=1}^{q} \beta_j \sigma_{t-j}^2 \qquad (3)$$

where,

σ_t^2 is the conditional variance

σ_{t-j}^2 is the past conditional variance

ε_{t-i}^2 is the past squared residual return

To forecast the values for stock returns of the firms, the GARCH model engages with all the distribution methods Normal, General Error and student's t distributions. As the Student's T distribution outperforms the Normal and General error Distribution in computing the future value of the stock returns for the financial time series, the study extensively builds the GARCH models in accordance with these distributions. Apart from using the proficient volatile GARCH(1,1) model, the analysis also computes the forecasting with the help of the asymmetric GJR-GARCH (Glosten Jagannathan Runkle-GARCH) model for ensuring efficient daily returns. While studying the volatility of the stock returns in Nigeria, authors concluded that the GJR-GARCH model proved to be the best predictive model for forecasting the stock returns in comparison to the other GARCH models (Onwukwe et al., 2011).

Before using the GARCH models to fit the dataset , one specific requirement of the analysis is to check the volatility of the data (stocks). This can be done by checking the kurtosis value *K* and skewness *S* of the stocks data after defining the distribution method for the analysis. Kurtosis and skewness are defined as the measure of peakness and symmetry for the distribution of the data ().The basis entity for a series of data being volatile is its kurtosis value being above 3 and skewness aligned either to the left or right (). Kurtosis *K* and Skewness *S* are expressed as follows :

$$K = \frac{1}{n} * \frac{\Sigma_{i=1}^{n} (y_i - \bar{y})^4}{\left(\frac{1}{n} \Sigma_{i=1}^{n} (y_i - \bar{y})^2\right)^2} \qquad (4)$$

$$S = \frac{1}{n} * \frac{\Sigma_{i=1}^{n}(y_i - \overline{y})^3}{\frac{1}{n}\Sigma_{i=1}^{n}(y_i - \overline{y})^{\frac{3}{2}}} \quad (5)$$

where ,

$\overline{y}$ is the mean of the time series in the dataset

y_i is the events or the time series values in the dataset

n is the total number of observations

3.3 Model Estimation

After defining the model, the next step is to fit the model for the stock' closing price datasets. Before fitting the model, identification of the model parameters are important (p and q), this can be done by plotting correlogram for the autocorrelation (ACF) and Partial autocorrelation(PACF) functions for the given training closing price datasets to regulate the autoregressive (AR) 'p' and moving averages (MA) 'q' orders. To estimate the parameters Maximum Likelihood Estimation (MLE) is used for the ARIMA and GARCH models to satisfy the properties of point estimation .

MLE for the ARIMA model is expressed as ,

$$l = -\frac{n}{2} ln\{2\pi \hat{y}_a^2\} - \frac{\Sigma_{t=1}^{n} e_t^2}{2\hat{y}_a^2} \quad (6)$$

where,

l is the log-likelihood function for the specified distribution

e_t are the error terms for the ARIMA model from eq.(1)

$\hat{y}_a = \frac{\Sigma_{t=1=p}^{n} e_t^2}{df}$, df = n - (2p + q +1)

MLE for the GARCH model can be expressed as ,

$$l = -\frac{n}{2}\ln(2\pi) - \frac{1}{2}\Sigma_{t=1}^{n}\{ ln(\sigma_t^2) + \frac{e_t^2}{\sigma_t^2}\} \quad (7)$$

where ,

l is the log-likelihood function for the specified distribution

e_t are the error terms for the GARCH model from eq.(3)

σ_t^2 is the conditional variance from eq.(3)

Next , to effectively find out the best fit models for the given dataset of stocks, Akaike Information Criterion (AIC) and Bayesian Information Criterion (BIC) are determined for ARIMA and GARCH models . Chakravarty and Ghosh (2011) , showed that AIC and BIC are the appropriate criteria to estimate the best fit and correct models used for predicting future values (Chakravarty and Ghosh,2011). The smaller the value for both criterions, the better the model is to fit the dataset. For ARIMA and GARCH models, the criterions are expressed as :

Akaike Information Criterion (AIC) ,

$$AIC = -2 \log(l) + 2*(p+q+1) \quad (8)$$

Bayesian Information Criterion (BIC) ,

$$BIC = -2 \log(l) + ((p+q+1)*\log (n)) \quad (9)$$

where,

l is the log-likelihood function for the distribution

p is the autoregressive (AR) component of ARIMA and GARCH models

q is the moving average (MA) component of ARIMA and GARCH models

3.4 Model Evaluation

After estimating the appropriate models , they are used to forecast the future values of the stock returns for each technical firm depending on the required time horizons. For this study, values are forecasted for two years depending on the type of data chosen for making investors' decision making even easier by gauging the future flow of stock movements, helping them make accurate investments in the technical firms. But before producing the appropriate model for this decision-making process, each model should be evaluated on the basis of their accuracy to forecast suitable values (Armstrong and Collopy, 1992). Accuracy of each model can be evaluated on the terms of the testing dataset by the forecast measure of Mean absolute Percentage Error (MAPE) and Root Mean Square Error (RMSE) . Models producing small values of MAPE and RMSE concur as the better model for the given closing price dataset of the stocks.

$$\text{MAPE} = \left\{ \left[\Sigma_{i=1}^{n} \left| \frac{y_i - \widehat{y}_i}{y_i} \right| \right] / n \right\} \times 100\% \tag{10}$$

$$\text{RMSE} = \sqrt{\{\Sigma_{i=1}^{n} (y_i - \widehat{y}_i)^2 / n} \tag{11}$$

where,

y_i is the actual value of the observations

$\widehat{y_i}$ is the forecasted value for the observations

n is the number of period of the observations

4. Analytical Results and Discussions

Before forecasting the models to predict the future stock return values for the technical firms, analysing the type of stock data used in building the models is important. Table 4.1 shows the descriptive statistics for the daily closing price stock data for the given companies from 2015 to 2022. Following the statistics, graphical representation of the stock movements of all the companies are portrayed in fig 4.1 for the given time horizon.

Stocks	Mean	Median	Max.	Min.	Std Deviation	Kurtosis	Skewness
BA.L	567.163	543.5	867	462	93.20701	1.527393	1.0636
SGE.L	660.6	672	853.8	459.8	81.87284	-0.37180	-0.35674
SN.L	1351.3	1313.5	1990	984.6	213.8064	0.840013	0.20316
CCC.L	1482.2	1256	3030	673.5	717.5727	-1.0413	0.63775
GNS.L	2723	2480	6070	1224	1185.518	0.21225	1.018325
QQ.L	278.433	277.55	394.2	181.1	48.274	-0.39658	0.07402
FLTR.L	8978.33	8453.36	16915	3794.394	2658.649	0.124051	0.777355
RWS.L	422.85	439	747	128	162.8075	-0.23379	-1.102352
SPT.L	165.6	130.20	304	65.25	76.44943	-1.544731	0.303191
SXS.L	2560	2558	4083	1442	517.1208	0.418121	0.176965

Table 4.1 Descriptive Statistics for the companies Closing Stock Price Data (20015-2022)

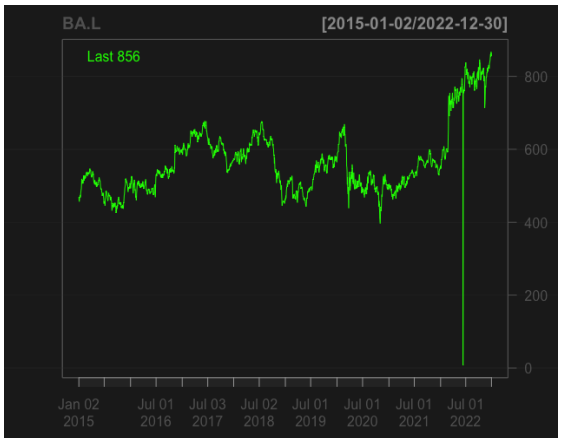


Fig 4.1 BAE Systems (BA.L)

Fig 4.2 The Sage Group PLC (SGE.L)

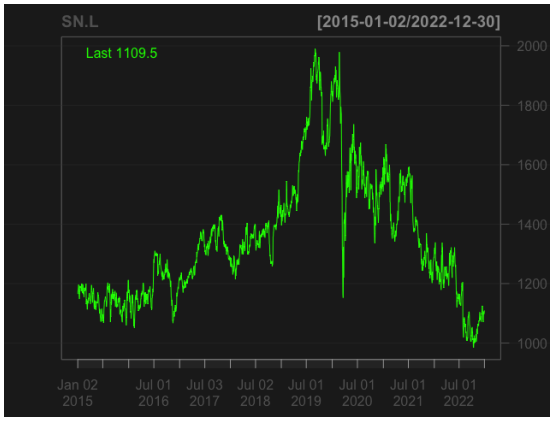


Fig 4.3 Smith and Nephew PLC (SN.L)

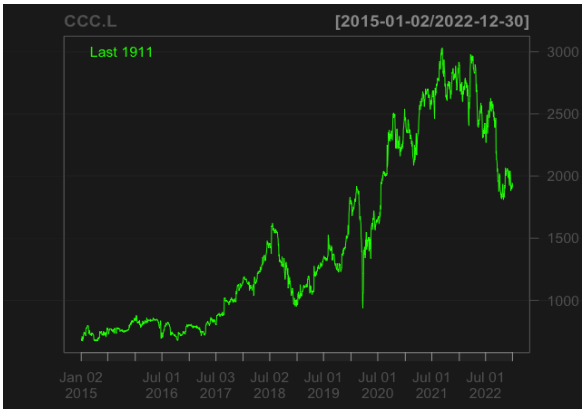


Fig 4.4 Computacenter PLC (CCC.L)

Fig 4.5 Genus PLC (GNS.L)

Fig 4.6 QinetiQ Group (QQ.L)

Fig 4.7 Flutter Entertainment (FLTR.L)

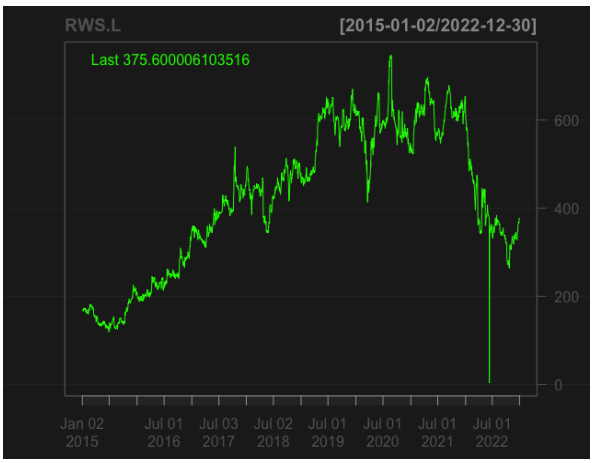


Fig 4.8 Renishaw PLC (RWS.L)

Fig 4.9 Spirent Communication PLC (SPT.L)

Fig 4.10 Spectris PLC (SXS.L)

As the data selected for the analysis are on the basis of daily observed stock movements, but for investors to get more subsidised views on their investment decisions, the models will also be efficient on the basis of weekly and monthly stock movements. Empirical analysis of the data are carried out on the concerns of ARIMA and GARCH models to show a comparative performance. But before evaluating both the model's efficiency, estimation of ARIMA and GARCH models holding specific orders is concerned. To confront this function and build the

models, both the methods govern an internal evaluation of models to specify the governing parameters to profoundly execute the operation of forecasting the stock values.

4.1 Model Estimations

Described from the visualisation above in the figures , the data is fabricated and fed to the model where the dataset is split into the training and testing samples as specified in fig 3.1. As we know to build an ARIMA model, we need to sustain a stationary financial time series data .This is done by adhering to the Augmented Dickey-Fuller (ADF) Test, where all the training datasets are trialled for the stationarity in their forms. Followed by the plots of autocorrelations (ACF) and Partial Autocorrelation functions (PACF). In the given context , for reference, if we choose the closing stock price dataset of the Smith and Nephew PLC (SN.L) , the dataset is considered to go through the ADF test where the residual p-value decides the stationarity of the data. In this case , the p-value produced for SN.L stocks dataset is 0.08821, which is higher than the adjusted p-value of 0.05, which rejects your hypothesis of data being a stationary time series. To convert the given dataset into a stationary time series, the dataset is differentiated numerous times to produce a stationary time series . In the context of "SN.L" stocks, only a single differencing is enough to convert the given stock movements into a stationary time series, generating the value of the differencing parameter as "1". Followed by this ACF and PACF plots are generated to identify the autoregressive (AR) and moving average (MA) parameters p and q to automate the orders of the ARIMA model(Schaffer et al.,2021).

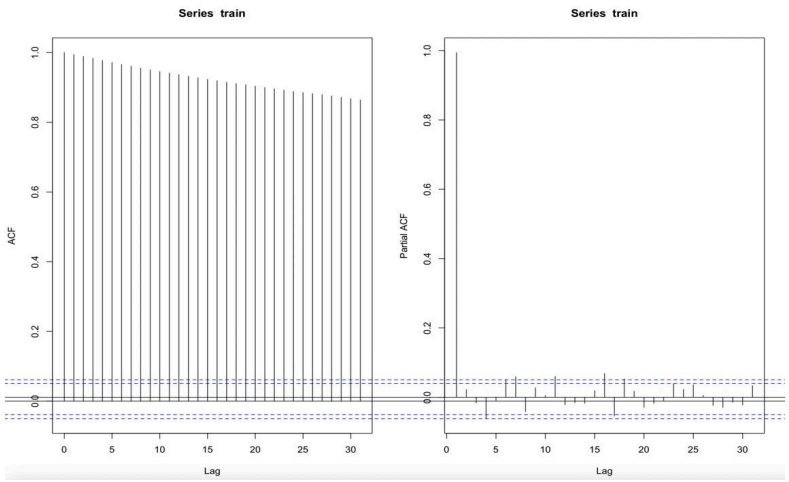
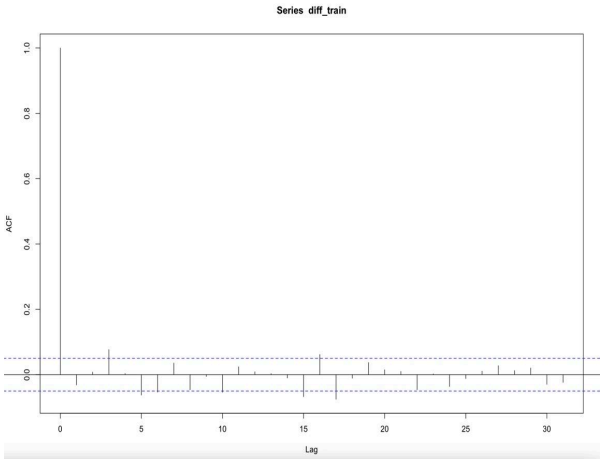


Fig 4.11 ACF and PACF plots for general SN.L daily stock movements



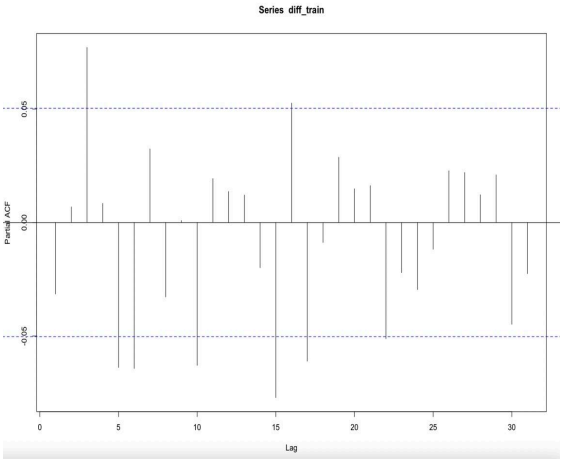


Fig 4.12 Differentiated ACF Plot of SN.L Stocks

Fig 4.13 Differentiated PACF Plot of SN.L Stocks

Depending on the above depicted figures , we can gauge the parameters of the AR and MA terms to extensively test the models for various orders to fit the given dataset of stock returns , helping the investors to specify a model to forecast the future values. To fit the model for a given dataset, the "*arima"* function is used to test the orders with different specifications, enabling a perfect blend with the stock movements of the firms, following an order of (p,d,q). The estimation of the order for the specification of the best fit model is usually governed by its omitting values of log-likelihood(l), Akaike Information Criteria(AIC) and Bayesian Information Criteria(BIC) (from eq.(7)(8)(9)) (Cavanaugh and Neath , 2019). Order (p,d,q) that

depicts low information criterion values and high log-likelihood values are considered to be the best fit models for the given datasets(Bozdogan,1987).

Arima Order (p,d,q)	Log-Likelihood (l)	AIC value	BIC value
(1,1,1)	-6880.06	13766.12	13794.71
(2,1,2)	-6870.13	13750.12	13783.01
(2,1,3)	-6868.56	13749.13	13781.07
(3,1,2)	-6868.53	13749.06	13781.02
(3,1,3)	-6868.51	13751.02	13782.28

Table 4.2 Order(p.d,q) Estimation for daily stocks movement of SN.L Stocks

From the above table we can say that ARIMA order (3,1,2) best fits the model to predict the future returns values of the Smith and Nephew PLC company (SN.L). The coefficients required to format this order into a model are shown below, with a raging equation of the model helping us predict the values of required horizons through the ARIMA model.

	AR1	AR2	AR3	MA1	MA2
Coefficients	0.2158	-0.7429	0.0542	-0.2495	0.7686

Table 4.3 Coefficients of Estimated order (3,1,2) for Smith and Nephew PLC (SN.L) Stocks

ARIMA model Equation for the stocks of SN.L with order (3,1,2) is ,

$$y_t = 0.2158\, y_{t-1} - 0.7429\, y_{t-2} + 0.0542\, y_{t-3} + \varepsilon_t - 0.2495\, \varepsilon_{t-1} + 0.7686\, \varepsilon_{t-2} \quad (12)$$

In a similar manner various technical companies taken under consideration to construct the ARIMA model are processed in the same entity of log likelihood and information criterion

values , to provide the order that best fits their respective datasets of stock movements .This orders of specific companies are listed below in Table 4.4 along with their identification values of criterions .

Company Name	ARIMA order (p,d,q)	Log-likelihood value (l)	AIC Value	BIC Value
The Sage Group PLC (SGE.L)	(0,1,1)	-5683.71	11371.41	11382.06
Smith and Nephew PLC (SN.L)	(3,1,2)	-6868.53	13749.06	13781.01
BAE Systems (BA.L)	(0,1,0)	-5279.25	10560.49	10565.82
Computacenter PLC (CCC.L)	(2,1,2)	-7195.33	14402.65	14434.6
Genus PLC (GNS.L)	(0,1,1)	-7942.26	15890.53	15906.5
QinetiQ Group (QQ.L)	(2,1,2)	-4383.76	8777.51	8804.13
Flutter Entertainment (FLTR.L)	(2,1,2)	-9931.2	19874.39	19906.34
Renishaw PLC (RWS.L)	(3,1,2)	-5612.27	11236.55	11268.49
Spirent Communication PLC (SPT.L)	(1,1,3)	-4241.46	8494.92	8526.87
Spectris PLC (SXS.L)	(1,1,0)	-7806.06	15616.12	15626.77

Table 4.4 Arima order for Technical firms along with there Criterion values

Computing the specific orders for each technical firm and constructing the ARIMA models on the basis of these orders , makes the models capable of forecasting the stock return values for the respective companies on the valuation of their presented past stock movements . These ARIMA models will practically help the investors to gauge the stock returns gains they will be acquiring from the technical firms in the mere future, assisting them in making accurate investment decisions on these stocks. As the orders specified for the stock movements are on their daily movement basis , if imputed to run the models on a weekly and monthly basis, the ARIMA orders of the companies may be subjected to change in accordance with the modification provided in movements of stocks dataset. Efficiency of these models are evaluated at later stages in comparison with the GARCH model making investors a coherent object to assist their investment decisions according to their required time horizon of predictions.

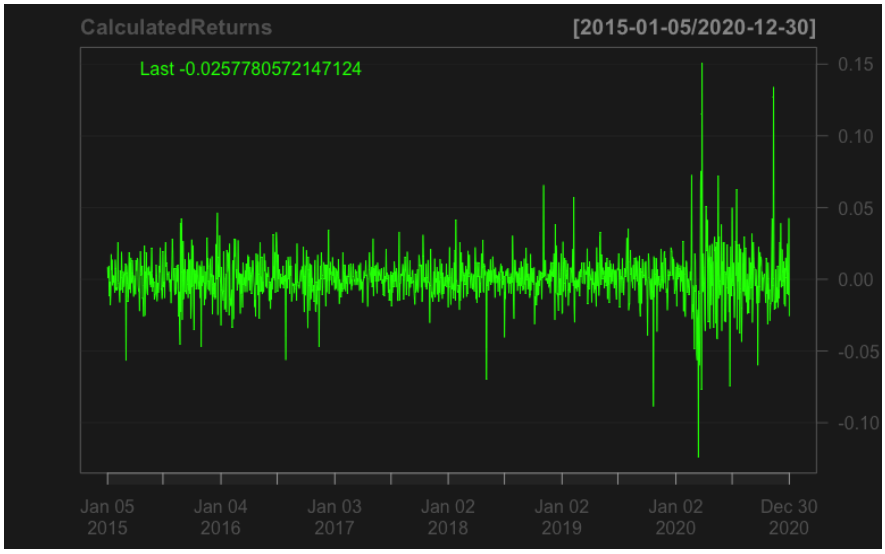


Fig 4.14 Calculated return values for the stock price data of SN.L stocks

Criterions	GARCH (1,1) with "norm" Distribution	GARCH(1,1) with "GED" Distribution	GARCH(1,1) with "sstd" Distribution
Goodness-of-Fit Test (p)	2.76×10^{-12}	0.01920	0.0998
Log-likelihood Value	4301.946	4412.003	4437.312
Akaike Information	-5.6664	-5.8102	-5.8435
Bayes Information	-5.6523	-5.7926	-5.8225
ARCH-LM Test (p)	0.5333	0.6723	0.6799

Table 4.5 Criterion values for Estimation of Symmetric GARCH Models with different distribution methods

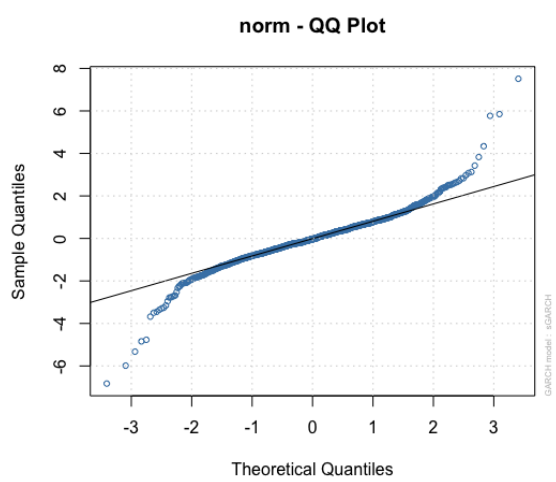


Fig 4.15 Symmetric Garch with "Normal " Distribution

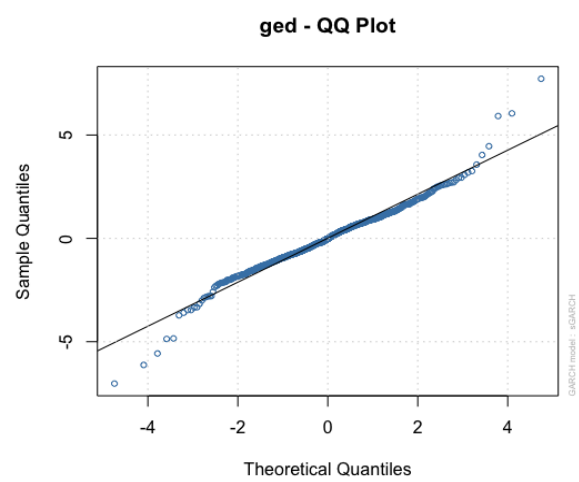


Fig 4.16 Symmetric Garch with "GED " Distribution

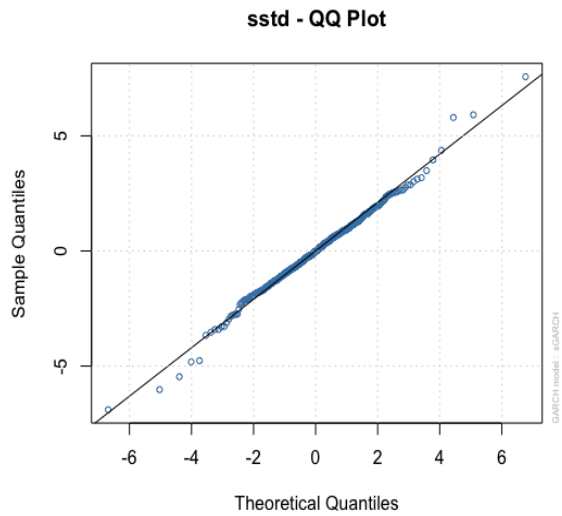


Fig 4.17 Symmetric Garch with "sstd " Distribution

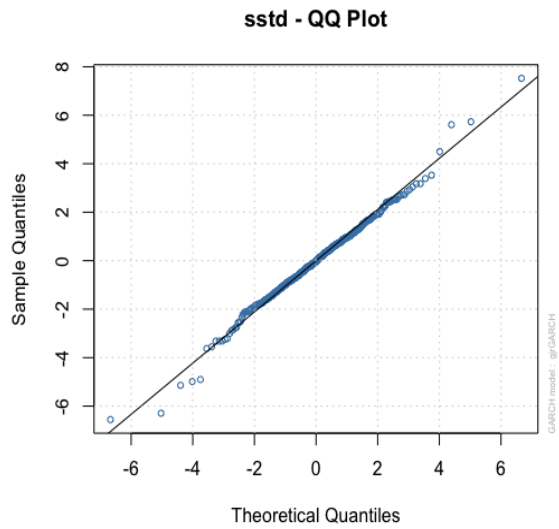


Fig 4.18 Asymmetric Garch with "sstd " Distribution

From the results above in figures and table 4.6 we can clearly gauge that among all the distribution models in the symmetric GARCH (1,1) , only student-t distribution fully satisfies the conditions of model estimations. As the other distribution failed in fitting the dataset by showing low value for the p-statistics in the Goodness-of-fit test. Considering the "sstd" distribution extremely fitting the stock movements dataset, these values are further analysed in comparison to the asymmetric GARCH model named GJR-GARCH(1,1) for the given stock returns movement to accurately forecast the future volatility of the stocks for respective technical firms. Table 4.6 below shows the analytical comparison of symmetric sGARCH(1,1) and asymmetric GJR-GARCH(1,1) models on the same biassing criterias of model estimation tests for SN.L stocks movement .

Criterions	Symmetric GARCH(1,1) models with "sstd" distribution.	Asymmetric GJR-GARCH(1,1) with "sstd" distribution .

Goodness-of-Fit Test (p)	0.0998	0.13299
Log-likelihood Value	4437.312	4439.586
Akaike Information	-5.8435	-5.4405
Bayes Information	-5.8225	-5.8293
ARCH-LM Test (p)	0.6799	0.6881

Table 4.6 Comparison of Symmetric and Asymmetric GARCH models

	μ	ω	α_1	β_1	γ_1	Skew	Shape
Coefficients	4.06×10^{-04}	2.6696	2.20×10^{-02}	0.951	2.94×10^{-02}	0.999	4.0939

Table 4.7 Coefficients of GJR-GARCH(1,1) Model for SN.L Stocks

GJR-GARCH(1,1) equation for SN.L is as follows ,

$$\sigma_t^2 = 2.6696 \times 10^{-6} + 5.153 \times 10^{-2}\, \varepsilon_{t-1}^2 + 0.9518\sigma_{t-1}^2 \qquad (13)$$

From table 4.6 we can evaluate that the asymmetric GJR-GARCH(1,1) model better fits the dataset of stock movement to effectively and accurately forecast the stock return prices of technical firms in comparison to the symmetric GARCH(1,1) models. As both the models influence a good fit for the stock movement datasets, showing the non-existence of correlation in the dataset by giving the value of ARCH-LM test above 0.05 . But by producing high log

likelihood and low information criterion values GJR-GARCH(1,1) model outperforms the symmetric model to efficiently fit the stock movement datasets of the SN.L technical firm. In the similar manner , after analysing all the datasets of various technical firms as shown in table 4.1 for all the above mentioned GARCH symmetric and asymmetric models , the estimations prove that GJR-GARCH(1,1) model effectively better fits the daily stock movements datasets to accurately forecast the future stock returns values in comparison to the symmetric GARCH(1,1) models. Efficiency of the GJR-GARCH(1,1) model is evaluated by simulating the forecast of the returns of stock movements and its volatility through the *ugarchforecast* function , for a required time horizon of two years(2021-2022) . The simulation of the models gives the fitted and sigma values for each time set horizon as shown in figures below , computing two returns for the same data observation. To accurately measure the forecast of the GJR-GARCH(1,1) model the produced fitted return values are converted into acute values of currency stocks to compare the results with their propagating ARIMA model to make accurate investment decisions.

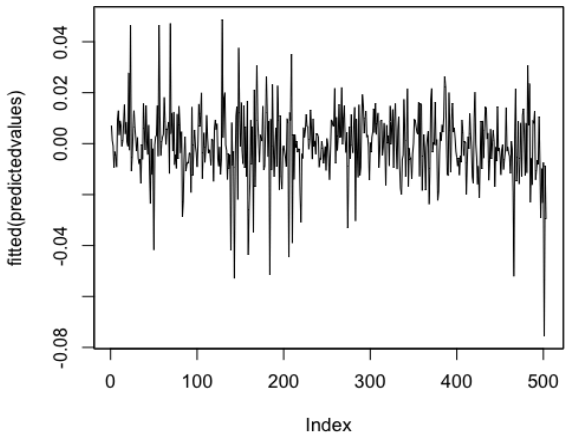


Fig 4.19 Forecasted Fitted GARCH model Values

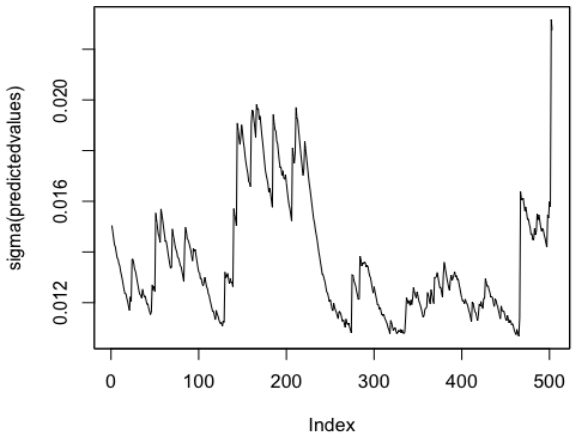


Fig 4.20 Forecasted Sigma GARCH model Values

4.2 Model Evaluation

Evaluating the model performance of both the ARIMA and GARCH models, resulting parameters are compared to provide accurate forecasting measures being the most efficient models for investors to make precise decisions on their investments for the technical firm and the UK Tech sector for the long run gain. The Forecasting measures are formulated for multi-step ahead forecasts of two years time horizon (01 Jan 2021 - 31 Dec 2022) , on the terms of Root Mean Square Error (RMSE) , Mean Absolute Percentage Error (MAPE) to best fit the model of stock movements datasets. Table 4.7 shows the measuring accuracy forecasts for ARIMA andGARCH models in concern to all the 10 technical firms .

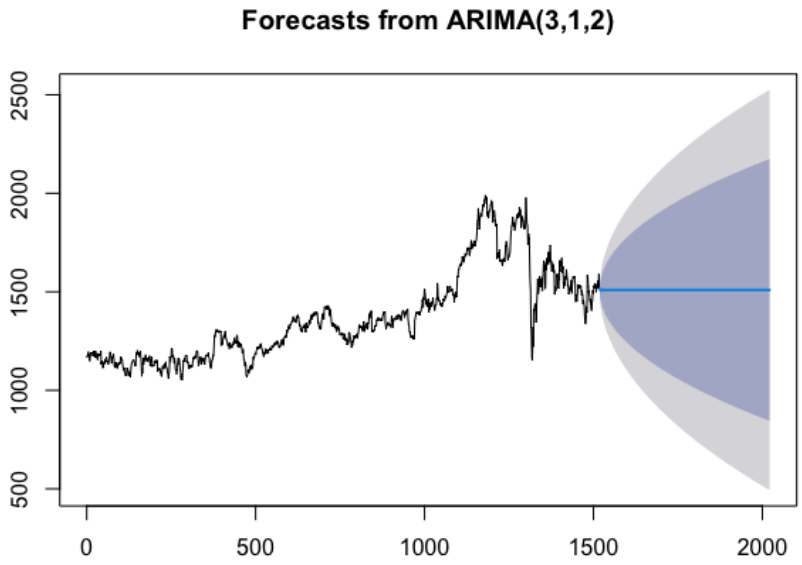


Fig 4.21 Actual Forecasted Stock return values for SN.L stocks through ARIMA(3,1,2) Model

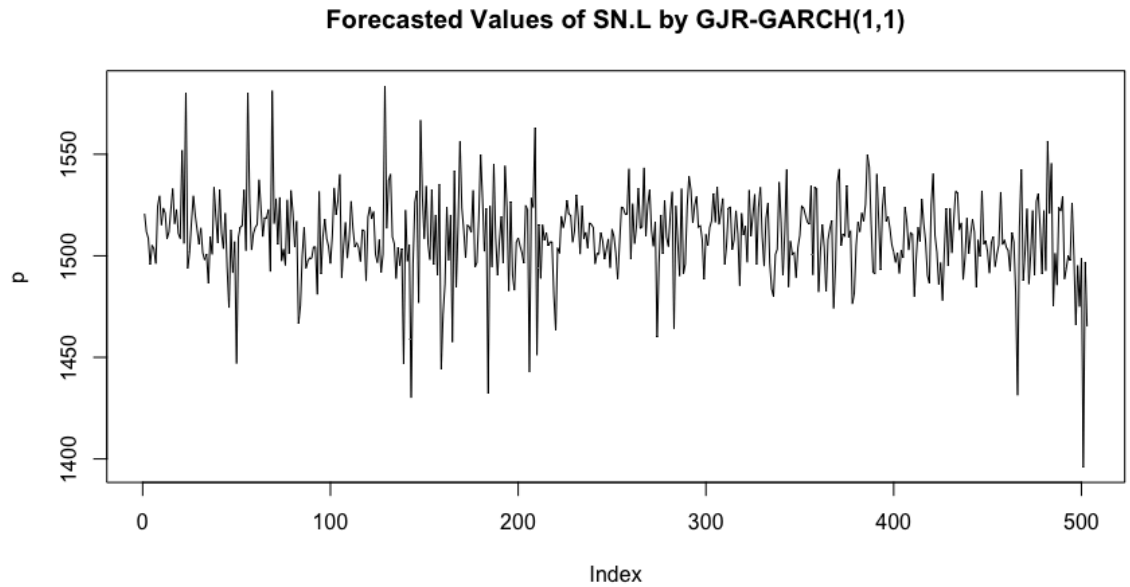


Fig 4.22 Actual Forecasted Stock return values for SN.L stocks through GJR-GARCH Model

Companies	Arima Models (p,d,q)		GARCH Models (p,q)	
	RMSE	MAPE(%)	RMSE	MAPE(%)
The Sage Group PLC (SGE.L)	10.2513	1.0876	7.0295	0.6624
Smith and Nephew PLC (SN.L)	22.3854	1.0547	19.023	0.5243
BAE Systems (BA.L)	7.8521	1.0536	5.3354	0.9528
Computacenter PLC (CCC.L)	27.7668	1.3099	27.8386	1.2955
Genus PLC (GNS.L)	45.4329	1.3024	35.5456	1.0004
QinetiQ Group (QQ.L)	4.3510	1.0623	22.9720	1.9371

Flutter Entertainment (FLTR.L)	168.563	1.4013	114.3916	1.0597
Renishaw PLC (RWS.L)	9.7850	1.6560	9.7316	1.6309
Spirent Communications PLC (SPT.L)	3.9616	1.5850	1.7772	1.1464
Spectris PLC (SXS.L)	41.562	1.2856	39.227	1.2846

Table 4.8 Accuracy Forecast Measure Comparison of ARIMA and GARCH Models

Evident from the above table that both the models perform absolutely well, producing low margin of percentage error . As the error percentage in the models are way below the margin of 5% both the models are considered producing accurate forecasts of stock returns for given firms dataset, forming a base to help the investors make accurate investment decisions on these companies. But considering the comparative performance of both the models, GJR-GARCH (1,1) outlashes the ARIMA models in predicting the volatility of stock returns value, making a better fit for the datasets for a long run assuming predictions of values. But in the near margin of short-term prediction of returns, the case might upturn towards the ARIMA model with precise order to efficiently outline the forecast of values. Out of the 10 considered firms, the asymmetric GARCH model performs better fit for the datasets of 9 firms giving low percentage value of MAPE and RMSE, except in the case of QinetiQ PLC , which is due to the variability of decreasing volatile impact on the stock movements . The residuals of these models hence follow a normal distribution to estimate the predictive forecast, but the asymmetric GARCH models provide smaller magnitude of values for the residuals in comparison to the corresponding ARIMA models to estimate the volatility of the considered financial time series making it a better fit model to forecast the stock returns of technical firms analysed in the UK FTSE techMark series.

5.Conclusion

As the results reported in association to the Stock Return Forecasting of Top UK technical firms show that the asymmetric GARCH models named GJR-GARCH(1,1) better fits the daily stock movements datasets in comparison to corresponding ARIMA models to forecast the stock return values, making a ease of pathway for investors to take accurate investment decisions on the companies listed in FTSE TechMark Series. The results align with findings of studies done by (Kambouroudis et al.,2015; Tamilselvan and Vali , 2016 ; Chand et al., 2012) . The results also report that the symmetric GARCH models also fit the dataset to forecast the returns but only through student-t type distributions (sstd), whereas the models with normal and general error distribution failed to possess the goodness-of-fit test. Further, in comparison among the GARCH models both the symmetric and the asymmetric models display the values greater than the 0.05 statistics for Goodness-of-fit and ARCH-LM test indicating non-existence of correlation among the data observation of stock movements . But the asymmetric GJR-GARCH(1,1) model emerges as the best fitted model depicting precise criterion values of high log-likelihood and low Akaike-Bayes Information criteria values. Provided as the best estimated fit for the stock movements , these results are tested for all the considered 10 technical firms datasets of daily stock movements, whose results conclude that GJR-GARCH(1,1) is estimated to be the best fit model for the given technical firms stock movements to forecast their respective stock return values.

ARIMA models on the other hand also indicate a good fit model for the considered daily stock movements datasets. As each dataset stock movements tend to operate with different ARIMA orders depending on their fluctuation of stock price, an internal estimation of order fit is carried out for every possible combination formed for individual datasets of technical firms. This estimation is carried out on the basis of criterion values depicted by each order (p,d,q) mentioned for a particular dataset. Order (p,d,q) having high log-likelihood and low information criterion values is considered as the best ARIMA order for that particular stock movement dataset. Governing this coefficient of the orders estimated by the ARIMA model are computed to form an individual ARIMA Equation. After estimating the ARIMA and GARCH models for every individual technical firm, these models are compared on the basis of accuracy forecast measures

to indicate the better fit forecasting model for predicting the stock return values. Asymmetric GJR-GARCH models indicating a low value of RMSE and MAPE forms a better fit in comparison to corresponding ARIMA models for 9 out of 10 technical firms mentioned in the FTSE techMark series . The standing observation of QinetiQ PLC company where ARIMA models work efficiently in comparison to GJR-GARCH(1,1) is due to the positive effect of decreasing volatility in the stock movements.

Hence on final evaluation of the models it is concluded that GJR-GARCH better fits the model in comparison to the ARIMA models to forecast the future stock returns value for technical firms assisting investors to make accurate decisions of investment in the UK tech sector. Governing the growth of the UK tech sector in the past years, investments in these sectors have played a crucial role in manipulating the financial situation and economic growth. Stock market on the other hand holds an important aspect in developing the companies falling under this sector, as many companies have registered their profiles publicly in the Tech Sector seeing the growth features in the FTSE techMark series . Making accurate investment in these technical chip stocks is more predominant considering its direct impact of stock market fluctuations. Hence , to assist these investment decisions with accuracy the established GJR-GARCH and ARIMA models can be used to gauge the long-term and short-term stock return values of companies falling under the list of FTSE techMark series in the UK Tech sectors.

Limitations

Even though the models are effective in predicting the stock return price but are adhered by some limitations, such as regime shifts experienced in the trends and markets of financial institutes due to unexpected events creating inaccuracies in the predictive models(Bloomfield and Hales, 2002). Manipulation market is also a hurdle for predictive models where external factors such as algorithmic trading can create unexpected fluctuations in the stock movements , making it difficult to forecast the accurate stock prices. Market sentiments and human emotions also play a crucial role in controlling the stock movements, where forecasting models usually fail to integrate the social media trends and news sentiment (Zhang et al.,2018). Forecasting of stocks is affected by the hypothesis governed by the publicly available information that continuously

reflects on the stock movements, making it challenging for the models to only rely on the past historical data (Madge and Bhatt,2015). This efficient hypothesis at times can turn the predictions down when the formations of human feelings and stock sentiments can manipulate the stocks which are out of the reach of forming functional models. But considering all the limitations the models are yet effective for long-term investment decisions , helping the investors to gauge the range of movements the stocks have manipulated. The model's efficiency with considered human sentiments will help the investors to make accurate investment decisions on the techMark stocks governed by the model fits of GJR-GARCH(1,1) and ARIMA models for long-term and short-term acquires.

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Appendix-1

Ethical Form

Internal research ethics application form

For module LUBS5579M (Business Analytics and Decision Sciences Dissertation) covered by University of Leeds ethical approval reference AREA 17-055[1].

Student ID	201650475
Student Name	ANIKET AMAR NERALI
Degree Programme	MSc Business Analytics and Decision Sciences
Title / topic area	Stock Return Forecast of Top Technical Firms: By ARIMA and GARCH Models
Name of dissertation supervisor	Dr. Wessam Abouarghoub

[1] Ethical approval valid until 31/12/2027.

Are you planning to work with (data on) human participants for your dissertation?	Please tick the relevant box
Yes (This includes interviews, surveys and secondary data analysis of social media or internet data).	✓
No, I am conducting an in-depth literature review with analysis.	

If you ticked 'No' you do not need to take further action in respect of ethical approval. Please proceed to the declarations on page 7.

If you ticked 'Yes' you need to complete the rest of this form.

You MUST discuss your research design and the ethical issues it raises with your dissertation supervisor and receive their signed approval before you approach any participants or collect any data.

You MUST include a copy of your ethics form (signed by your supervisor) as an appendix to your final dissertation submission (both the electronic and paper copies).

INTERNAL RESEARCH ETHICS APPLICATION
Part A: Compliance with the module's block ethical approval

Ethical review is required for **all research involving human participants**. Further details of the University of Leeds ethical review requirements are provided in the *Research Ethics Policy* available at:
http://ris.leeds.ac.uk/ResearchEthicsPolicies and www.leeds.ac.uk/ethics.

1. **Will your dissertation involve any of the following?**	**Yes**	**No**
New data collected by administering interviews for analysis		✓
New data collected by administering questionnaires for analysis		✓
New data collected from observing individuals or populations		✓
Working with secondary aggregated or population data		✓
Using open data	✓	
Using data given to you by a company		✓
Any other research methodology, please specify:		✓

2. **Will any of the participants be from any of the following groups?** (Tick as appropriate)	**Yes**	**No**
Children under 16		✓
Adults with learning disabilities		✓
Adults with other forms of mental incapacity or mental illness		✓
Adults in emergency situations		✓
Prisoners or young offenders		✓
Those who could be considered to have a particularly dependent relationship with the investigator, e.g. members of staff, students		✓

Other vulnerable groups, please specify:		✓

3. **Will the project/dissertation/fieldwork involve any of the following:** (You may select more than one)	**Yes**	**No**
Patients and users of the NHS (including NHS patients treated under contracts with private sector)		✓
Individuals identified as potential participants because of their status as relatives or carers of patients and users of the NHS		✓
The use of, or potential access to, NHS premises or facilities		✓
NHS staff - recruited as potential research participants by virtue of their professional role		✓
A prison or a young offender institution in England and Wales (and is health related)		✓

If you have answered 'Yes' to **ANY of the above statements in questions 2 or 3** then you will need to apply for full ethical approval which is a faculty committee level process. This can take up to 6-8 weeks, so it is important that you consult further with your supervisor and/or program director for guidance with this application as soon as possible. Please now complete and sign the final page of this document. The application form for full ethical review and further information about the process are available at http://ris.leeds.ac.uk/uolethicsapplication.

If you answered **'No' to ALL of the statements in Questions 2 and 3** please continue to part B.

INTERNAL RESEARCH ETHICS APPLICATION
Part B: Ethical considerations within block ethical approval

4. **Will the research touch on sensitive topics or raise other challenges?**	**Yes**	**No**
Will the study require the cooperation of a gatekeeper for initial access to groups or individuals who are taking part in the study (eg students at school, members of self-help groups, residents of a nursing home)?		✓
Will participants be taking part in the research without their knowledge and consent (eg covert observation of people in non-public places)?		✓
Will the study involve discussion of sensitive topics (eg sexual activity, drug use)?		✓
Could the study induce psychological stress or anxiety or cause harm or have negative consequences beyond the risks encountered in normal life?		✓
Are there any potential conflicts of interest?		✓
Does any relationship exist between the researcher(s) and the participant(s), other than that required by the activities associated with the project (e.g., fellow students, staff, etc)?		✓
Does the research involve any risks to the researchers themselves, or individuals not directly involved in the research?		✓

If you have answered 'Yes' to any of the statements above please describe the ethical issues raised and your plans to resolve them on a separate page. Agree this with your supervisor and submit it with this form.
You MAY be referred for light touch or full ethical review.

5. **International Research** Does your research involve participants outside of the UK?	**Yes**	**No**
Are any of your research participants located outside of the UK? For example: will you be gathering data through Skype interviews with participants located overseas?		✓
Will any of the fieldwork or research require you to travel outside of the UK to collect data?		✓

If you have answered 'Yes' to any of the statements above please describe the ethical issues raised with: gaining consent and gathering data from participants located overseas, securely storing and transferring data from the field back to the UK, any cultural issues that may be relevant. Please outline your plans to resolve this on a separate page and ensure that you have completed a risk assessment form (available from LUBS student education). Agree this with your supervisor and submit it with this form.

You MAY be referred for a light touch or full ethical review if you are unable to demonstrate that you have resolved the ethical issues relating to international research.

6. Personal safety Where will any fieldwork/ interviews/ focus groups take place?	**Yes**	**No**
At the university or other public place (please specify below).		✓
At my home address		✓
At the research subject's home address		✓
Some other location (please specify below).		✓

If you conduct fieldwork anywhere except at the university or other public place you need to review security issues with your supervisor and have them confirmed by the Module Leader who may refer you for a light touch or full ethical review. Write a brief statement indicating any security/personal safety issues arising for you and/or for your participants, explaining how these will be managed. Agree this with your supervisor and submit it with this form. Please note that conducting fieldwork at the research subject's home address will require strong justification and is generally not encouraged.

7. Anonymity	**Yes**	**No**
Is there any potential for data to be traced back to individuals or organisations, for instance because it has been anonymised in such a way that there remains risk? for example: highlighting people's positions within an organisation, which may reveal them		✓

If you have answered 'Yes' to question 7, please discuss this further with your supervisor. You need to provide a strong justification for this decision on a separate

sheet. This application will need to be reviewed by the dissertation Module Leader and may require a full ethical review.

8. Data management issues

Will the research involve any of the following activities at any stage (including identification of potential research participants)?		**Yes**	**No**
a.	Examination of personal records by those who would not normally have access		✓
b.	Sharing data with other organisations		✓
c.	Use of personal addresses, postcodes, faxes, e-mails or telephone numbers		✓
d.	Publication of direct quotations from respondents		✓
e.	Publication of data that might allow identification of individuals to be identified		✓
f.	Use of audio/visual recording devices		✓
g.	Storage of personal data on any of the following:		✓
	FLASH memory or other portable storage devices (e.g. USB storage)		✓
	Home or other personal computers		✓

	Private company computers		✓
	Laptop computers		✓

If you have answered 'Yes' to any of the questions above you must ensure that you follow the University of Leeds Information Protection Policy: http://www.leeds.ac.uk/informationsecurity and the Research Data Management Policy: http://library.leeds.ac.uk/research-data-policies#activate-tab1_university_research_data_policy.

You are obliged to provide a copy of your anonymised data to your supervisor for their records and to destroy other copies of your data when your degree has been confirmed.

Dissertation Research Ethical Approval: Declaration

For students	*Please tick as appropriate*
Option 1: I **will NOT** conduct fieldwork with (data on) human participants for my dissertation.	✓
Option 2: I **will** conduct fieldwork with (data on) human participants for my dissertation.	

For **options 1 and 2** - I confirm that:

- The research ethics form is accurate to the best of my knowledge.
- I have consulted the University of Leeds Research Ethics Policy available at http://ris.leeds.ac.uk/ResearchEthicsPolicies.
- I understand that ethical approval will only apply to the project I have outlined in this application and that I will need to re-apply, should my plans change substantially.

For **option 2** only:

- I am aware of the University of Leeds protocols for ethical research, in particular in respect to protocols on **informed consent**, **verbal consent**, **reimbursement for participants and low risk observation**. If any are applicable to me, signing this form confirms that I will carry out my work in accordance with them. http://ris.leeds.ac.uk/PlanningResearch

Student's signature: Aniket Amar Nerali (……..……..

Date: 06/07/2023
……..……..……..

For supervisors	**Yes**	**No**
No further action required		
I confirm that the dissertation is in line with the module's block ethical approval (Part A & question 8).	x	
I have discussed the ethical issues arising from the research with the student and agree that these have been accurately and fully addressed.	x	
I have reviewed the student's research proposal.	x	
I have reviewed the student's Risk Assessment Form (*if necessary*).		
Further actions required		

Refer to dissertation Module Leader for further review / discussion.		
The dissertation falls outside the module's block ethical approval and the student was advised to apply for full ethical review.		

Supervisor's signature: Wessam Abouarghoub.....…….……..

Date: 30/08/2023…….…….....……..

Appendix-2

Codes

```
#ARIMA Model
install.packages("quantmod")
library(quantmod)
# Input the Stock Variables for required firms predective analysis (SGE.L,SN.L,BA.L,CCC.L,GNS.L,QQ.L,FLTR.L,RWS.L,SPT.L,SXS.L)
data<-getSymbols("SN.L", src = "yahoo",from=as.Date("2015-01-01"),to=as.Date("2022-12-31"),auto.assign = FALSE)
#Daily Data
data
View(data)
#Used For forecasting Monthly data
#data<-to.monthly(data)
#Used For forecasting Weekly data
#data<-to.weekly(data)
# Construct only the closing price of stocks for forecasting
CLOSEPRICE<-data[,4]
CLOSEPRICE<-na.omit(CLOSEPRICE)
View(CLOSEPRICE)
summary(CLOSEPRICE)
sd(CLOSEPRICE)
skewness(CLOSEPRICE)
kurtosis(CLOSEPRICE)
sum(!complete.cases(CLOSEPRICE))
View(CLOSEPRICE)
#Visulise the data
chart_Series(CLOSEPRICE, col = "black")
add_SMA(n = 100, on = 1, col = "red")
add_SMA(n = 20, on = 1, col = "black")
add_RSI(n = 14, maType = "SMA")
add_BBands(n = 20, maType = "SMA", sd = 1, on = -1)
add_MACD(fast = 12, slow = 25, signal = 9, maType = "SMA", histogram = TRUE)
```

```
View(CLOSEPRICE)
# Set the training (JAN 2015- DEC 2020) and testing (JAN 2021-DEC 2022) dataset
set.seed(123)
train<-CLOSEPRICE[1:1518]
testa<-CLOSEPRICE[1519:2021]
plot(CLOSEPRICE,main="STOCK CLOSE PRICE,2015-2022",ylab="Price",xlab="Days")
lines(train,col="blue")
lines(testa,col="green")
legend("bottomright",col=c("blue","green"),lty=1,legend=c("Training","Testing"))
# ADF Test for Stationarity
library(tseries)
adf<-adf.test(train)
adf
#ACF and PACF Plots
acf(train)
pacf(train)
diff_train<-diff(train)
diff_train<-diff_train[-1,]
adf.test(diff_train)
par(mfrow=c(1,1))
acf(diff_train)
pacf(diff_train)
library(caTools)
library(forecast)
#Fitting the ARIMA Model
fitA<-arima(train,order=c(2,1,2))
fitA
tsdisplay(residuals(fitA),lag.max = 40)
checkresiduals(fitA)
fitb<-arima(train,order  = c(3,1,2))
fitb
BIC(fitb)
tsdisplay(residuals(fitb),lag.max = 40)
checkresiduals(fitb)
fitc<-arima(train,order = c(3,1,2))
fitc
tsdisplay(residuals(fitc),lag.max = 40)
checkresiduals(fitc)
fitd<-arima(train,order=c(2,1,2))
```

```
fitd
tsdisplay(residuals(fitd),lag.max = 40)
checkresiduals(fitd)
fite<-arima(train,order=c(2,1,2))
fite
tsdisplay(residuals(fite),lag.max = 40)
checkresiduals(fite)
#Forecasting the future values for stock returns
forecasta<-forecast(fitA,h=503)
plot(forecasta)
forecasta
forecastb<-forecast(fitb,h=503)
plot(forecastb)
forecastb
forecastc<-forecast(fitc,h=503)
plot(forecastc)
forecastc
forecastd<-forecast(fitd,h=503)
plot(forecastd)
forecaste<-forecast(fite,h=503)
plot(forecaste)
# Evaluate  the best ARIMA model for forecasting on the basis of forecast accuracy measures
accuracy(forecasta)
accuracy(forecastb)
accuracy(forecastc)
accuracy(forecasta)
accuracy(forecasta)
#Estimate the performance of the best ARIMA model with the available testing model
accuracy(forecasta,testa)


#GARCH MODEL
# Libraries
library(quantmod)
library(xts)
library(PerformanceAnalytics)
library(rugarch)
library(tseries)
```

```
# # Input the Stock Variables for required firms predective analysis (SGE.L,SN.L,BA.L,CCC.L,GNS.L,QQ.L,FLTR.L,RNS.L,SPT.L,SXS.L)
datag<-getSymbols("SN.L" ,src = "yahoo",from=as.Date("2015-01-01"),to=as.Date("2022-12-31"),auto.assign = FALSE)
#For Daily Data
View(datag)
0
#Used For Monthly data
#datag<-to.monthly(datag)
#Used For Weekly data
#datag<-to.weekly(datag)
# Construct only the closing price of stocks for forecasting
datag<-datag[,4]
CLOSEPRICE<-datag
CLOSEPRICET<-CLOSEPRICE[1:1518]
CLOSEPRICEF<-CLOSEPRICE[1519:2021]
chartSeries(SXS.L)
View(CLOSEPRICE)


# Calculate Daily returns
returnss <- CalculateReturns(CLOSEPRICE)
View(returnss)
returnss <- returnss[-1]
#Contruct the training (JAN 2015- DEC 2020) and testing (JAN 2021- DEC 2022) dataset
return<-returnss[1:1517]
test<-returnss[1518:2020]
class(test)
hist(return)
#Visualise the data
chart.Histogram(return,
        methods = c('add.density', 'add.normal'),
        colorset = c('blue', 'green', 'red'))
chartSeries(CalculatedReturns)

# Test for different Garch models
# 1. sGARCH model with contant mean
s <- ugarchspec(mean.model = list(armaOrder = c(0,0)),
        variance.model = list(model = "sGARCH"),
```

```
            distribution.model = 'norm')
m <- ugarchfit(data = return, spec = s)
coef(m)
plot(m)

9
0
# 2. GARCH with sstd
s <- ugarchspec(mean.model = list(armaOrder = c(0,0)),
            variance.model = list(model = "sGARCH"),
            distribution.model = 'sstd')
m <- ugarchfit(data = return, spec = s)
m
checkresiduals(m)
s <-m@fit$fitted.values
kurtosis(s)
plot(m)
9
0
# GARCH with General Error Distrubution
s <- ugarchspec(mean.model = list(armaOrder = c(0,0)),
            variance.model = list(model = "sGARCH"),
            distribution.model = 'ged')
m <- ugarchfit(data = return, spec = s)
m
plot(m)
9
0
# 3. GJR-GARCH
s <- ugarchspec(mean.model = list(armaOrder = c(0,0)),
            variance.model = list(model = "gjrGARCH"),
            distribution.model = 'sstd')
m <- ugarchfit(data = return, spec = s)
coef(m)
checkresiduals(m)
s <-m@fit$fitted.values
kurtosis(s)
plot(m)
9
```

```
0

#4. GJR-GARCH in mean
s <- ugarchspec(mean.model = list(armaOrder = c(0,0)),
            variance.model = list(model = "sGARCH"),
            distribution.model = 'ged')
m <- ugarchfit(data = return, spec = s)
m
plot(m)
9
0
# Simulation with the best GARCH model  evaluted on the bassis of providing the lowest Akakie
s <- ugarchspec(mean.model = list(armaOrder = c(0,0)),
            variance.model = list(model = "gjrGARCH"),
            distribution.model = 'sstd')
m <- ugarchfit(data = return, spec = s)
m
plot(m)
9
0
s
sfinal <- s
setfixed(sfinal)<-as.list(coef(m))
# Forecast the futre returns of the stocks by the GJR-GARCH model( Values for forecasting
changes according to the type of dataset choosed)
f <- ugarchforecast(m,
                n.ahead =503)

f
plot(f)
4
0
par(mfrow = c(1,1))
#Showing the volatality of the stock
plot(sigma(f))
predictedvalues<- ugarchpath(spec = sfinal,
              m.sim = 1,
              n.sim = 503,
              rseed = 123)
```

```
plot.zoo(fitted(predictedvalues))
View(fitted(sim))
class(test)
sigma(sim)
plot.zoo(sigma(predictedvalues))
tail(CLOSEPRICET)
# predicted values of stock return (Closing Price) of the firms on each day
#Substitute the last value of Stock close price from the trainiing set in the p variable to predict the future values from the GARCH model
p <- 1510*apply(fitted(predictedvalues),1, 'cumsum') +1510
p
View(p)
plot(p)
table(p)
class(p)

ForecastedValues
plot(p, type = "l", lwd = 1, main = "Forecasted Values of SN.L by GJR-GARCH(1,1)")
plot(p,closepricef, type = "l",lwd = 1)
length(p)
length(closepricef)
# Evaluate the Performance of GJR-Garch Model ,
View(p)
p<-as.numeric(p)
closepricef<-as.numeric(CLOSEPRICEF$SN.L.Close)
accuracy(p,closepricef)
p-closepricef

rmse<-sqrt(mean((p)^2))
rmse
abs<-abs((closepricef-p)/closepricef)*100

mean(abs)
forecastedp<-ts(p,start=1)
closepricets<-ts(CLOSEPRICEF$GNS.L.Close,start=1)
accuracy(forecastedp)
accuracy(forescastdp)
```

Evaluate the supreme model among ARIMA and GARCH model for being a better fit for the data depending on required time horizon

```
#ARIMA model
accuracy(forecasta,testa)
#GARCH model
accuracy(forecastedp,closepricets)
```